

Leopoldo Pla Sempere

Senior Research & Machine Learning Engineer · NLP · LLM Data · Distributed Systems

✉ b64encoded:bGVvcG9sZG9wbGFzZW1wZXJlQGdtYWlsLmNvbQ== | 🏠 lpla.github.io | 📄 github.com/lpla

Research and Machine Learning Engineer with 11+ years building NLP, data and production systems. **Led multilingual web-data pipelines processing >1 PB**, contributed to NLP/search serving **>7M queries/day**, and built Transformer-based research systems and open-source language technology. Strong in **Python/C++**, **multilingual NLP**, **distributed/HPC computing**, **data engineering** and **technical leadership**.

Experience

Senior NLP and ML Engineer, [Treelogic](#), Remote

Sep 2024 – Present

- Build applied-AI and data-platform components for **CARMEL** and **CRYPTOTRACK**, including authenticated **Trino/Delta Lake** backend access and integration with Airflow, Spark and MinIO.
- Develop shared Python infrastructure for **S3 ingestion and scraping**: paginated discovery, typed models, validation, date filtering, secrets/configuration, logging and automated tests.

Part-time Lecturer, [University of Alicante](#)

Oct 2019 – Present

- Teach programming, data structures, algorithms and software engineering to **200+ students annually** across Computer Science, AI and related degrees; design applied assignments and automated assessment.

Innovation Projects Engineer, [Grupo ENERCOOP](#)

Nov 2023 – Sep 2024

- Led technical work in national/EU R&I projects on AI, data and energy systems (DIGIRED-AI, InEEsX, MODERATE, STREAM, COMMUNITAS).

Senior NLP and ML Engineer, [University of Alicante](#)

Nov 2017 – Nov 2023

- Led **ParaCrawl/MaCoCu** engineering: multilingual crawling, extraction, language identification, alignment and filtering pipelines processing **>1 PB** of Common Crawl/WARC data on HPC/cloud infrastructure.
- Outputs included public monolingual/parallel corpora and data used to train language models.
- Led **MultiScore** engineering for Transformer-based optical and audio music transcription.
- Managed teams of **7+ engineers** and collaborated with international research partners including Johns Hopkins and the University of Edinburgh.

Data Scientist / NLP & ML Engineer, [Prompsit Language Engineering](#)

May 2015 – Oct 2017

- Contributed to data/retrieval infrastructure behind **Reverso Context**, a multilingual NLP/search product serving **>7M queries/day**, improving retrieval over large aligned corpora.

Research and Open Source

Bitextor — Core developer and maintainer of open-source multilingual web-data harvesting, extraction, alignment and cleaning for single-node to distributed/HPC environments.

CrossPoint Reader / Simulator — Maintainer working in C++ on EPUB rendering, constrained-memory reliability, parsers, image decoding, simulation and regression testing.

Selected publications — ParaCrawl: Web-Scale Acquisition of Parallel Corpora (ACL 2020); Building Domain-specific Corpora from the Web (BUCC 2022); MaCoCu corpus papers (EAMT 2022, 2023).

Talks — DockerCon 2022 on Bitextor and web-scale multilingual-data processing.

Technical expertise

ML & NLP — PyTorch, Hugging Face Transformers, multilingual NLP, information retrieval, data curation, model evaluation

Data & Distributed Systems — Common Crawl/WARC, Snakemake, Spark, Trino, Delta Lake, S3/MinIO, Airflow, HPC/cloud

Software Engineering — Python, C++, C, Java, Docker, Linux, Git, CMake, FastAPI, PostgreSQL, Solr

Languages — Spanish/Catalan native · English professional proficiency

Education

Polytechnic University of Valencia – M.Sc., Artificial Intelligence

Sep 2014 – Jun 2015

Specialized in Natural Language Processing and Deep Learning, thesis on audio-based classical composer identification.

University of Alicante – B.Sc., Computer Science

Sep 2010 – Jun 2014

Extraordinary Degree Award / Best Academic Record; B.Sc. thesis awarded highest honors.